Project Based Learning (PBL) – Machine Learning

Two Regression Problem Statements

Date: 27 October 2026

At a glance

Each project asks students to build a **new model**, train it on a **public dataset**, compare it with standard **baseline models**, explain it with **XAI**, and deploy it on the **NVIDIA Jetson Orin Nano**. Both projects are regression problems, and each student selects one of them.

No.	Project title	Regression task	Public dataset(s)
1	**SpikeDistill-RUL:** Teaching a Brain-Inspired Spiking Network to Predict Engine Failure	Time-series regression: remaining useful life (cycles)	NASA C-MAPSS turbofan engine degradation
2	**WD-GPR-SR:** When Uncertainty Meets Equations in Battery Health Prediction	Regression with uncertainty: battery state of health (%)	NASA Li-ion battery ageing data; CALCE battery data

Together, the two projects cover all ten techniques. The model names (SpikeDistill-RUL and WD-GPR-SR) are working titles for the models students will design. Each proposed model combines existing building blocks in a new way, so students must still check recent literature to confirm the novelty of their final design and cite the closest prior work.

Problem Statement 2 — WD-GPR-SR: When Uncertainty Meets Equations in Battery Health Prediction

Estimating the State of Health (SOH) of lithium-ion batteries is essential for electric vehicles, energy storage and portable electronics, because it indicates how much usable capacity remains and when a battery should be replaced. Gaussian Process Regression (GPR) is well suited to this task: it performs well on small datasets and provides a confidence interval with every prediction. However, standard GPR kernels operating on raw features struggle with complex, non-linear degradation, exact GPR scales poorly with data size, and deep-kernel variants can overfit and lose interpretability. Deep and wide neural networks and Bidirectional LSTMs (BiLSTM) capture non-linear patterns but provide neither calibrated uncertainty nor an explicit equation. Symbolic Regression (SR) can discover closed-form equations that engineers can read and verify, but it is sensitive to noise and rarely matches the accuracy of black-box models on its own. As a result, practitioners must choose between accurate, uncertainty-aware and interpretable models instead of obtaining all three.

In this research, the working principles and mathematical foundations of GPR (kernels, marginal likelihood, sparse approximations), Wide Neural Networks (WideNN), Deep Neural Networks (DeepNN), BiLSTM and SR will first be investigated using publicly available battery ageing datasets from NASA Ames Prognostics Center of Excellence and CALCE. Health indicators will be extracted from charge and discharge cycles (for example, constant-current charge time, voltage-plateau duration, incremental-capacity peak features and temperature rise), and baseline models will be evaluated using MAE, MSE, RMSE, R^2 and MAPE. Subsequently, a new model, **WD-GPR-SR**, will be developed. It uses a Wide-and-Deep kernel in which a wide linear branch and a deep neural feature extractor jointly define the GPR covariance, with all parameters learned through the marginal likelihood. A symbolic regression stage then discovers a compact closed-form correction for the GPR residuals, using a fitness function weighted by the GPR's predictive uncertainty. Explanations will be produced with SHAP, Automatic Relevance Determination (ARD) length-scales, PDP/ICE plots and the discovered equation itself, and the final model will be deployed on the NVIDIA Jetson Orin Nano for on-board battery monitoring. The proposed modifications should be supported by mathematical formulations, theoretical analysis or derivations wherever applicable, ablation studies and experimental validation.

Problem Statement 2:

To investigate the working principles, mathematical foundations and limitations of Gaussian Process Regression, Wide and Deep Neural Networks, BiLSTM and Symbolic Regression for lithium-ion battery state-of-health estimation using publicly available NASA and CALCE battery datasets, and to develop a new Wide-and-Deep-kernel Gaussian Process model with uncertainty-weighted symbolic residual correction through novel mathematical or algorithmic modifications. The proposed approach should address the limited expressiveness of standard GPR kernels, the overfitting and opacity of deep kernels, the absence of uncertainty in neural networks, the noise sensitivity of symbolic regression, and poor generalization to unseen battery cells. The proposed model should be validated through comparative analysis and ablation studies against conventional GPR, neural, recurrent, tree-based and symbolic baselines, and its effectiveness demonstrated in terms of predictive accuracy (MAE, MSE, RMSE, R^2 and MAPE), uncertainty quality (negative log-likelihood, prediction-interval coverage and width), equation complexity, explanation consistency, generalization to unseen cells, and inference latency, memory and energy per inference on the NVIDIA Jetson Orin Nano, with particular emphasis on accurate, trustworthy and interpretable battery health monitoring.

Item	Details
Proposed model	WD-GPR-SR: Wide-and-Deep kernel GPR (sparse variational for scale) plus an uncertainty-weighted symbolic residual equation
Baselines	GPR (RBF, Matérn, ARD), sparse GPR, standard deep-kernel GPR, SVR, Random Forest, XGBoost, WideNN, DeepNN, Wide & Deep NN, BiLSTM on cycle sequences, standalone SR (PySR or gplearn)
Datasets	NASA Ames battery ageing data (e.g. cells B0005, B0006, B0007, B0018); CALCE CS2 / CX2 cells
Evaluation protocol	Leave-one-cell-out cross-validation to test generalization to unseen batteries; fixed random seed
Metrics	MAE, MSE, RMSE, R^2, MAPE; NLL, 95% interval coverage (PICP) and width (MPIW); equation node count; latency, memory, energy on Jetson
XAI (at least 5)	SHAP, LIME, PDP, ICE, ARD length-scale relevance, symbolic equation sensitivities $\partial r/\partial x$
Edge AI	Export the deep feature extractor to TensorRT, store sparse-GP inducing points as arrays, compile the equation to C; measure each part on Jetson Orin Nano

Common rules, submission and evaluation (both problem statements)

Rules for every project

- **Public data only.** Use the dataset's official train/test split where it exists. Otherwise, create a fixed split with a stated random seed (e.g. `random_state = 42`) and use it for every model.
- **No data leakage.** Test data must not be used for training, hyperparameter tuning, model selection or threshold choice; use the validation set for these. Time-series data must be split chronologically.
- **Fair baselines.** All baselines and the proposed model use the same data pipeline, preprocessing, compute budget and seed. Where the problem statement asks for it, parameter budgets must also be matched.
- **Statistical reliability.** Report the final results as mean $\pm$ standard deviation over at least 3 random seeds.
- **Ablation study.** Remove or replace one component of the proposed model at a time to show what each part contributes.
- **Edge measurement.** On the Jetson Orin Nano, report the median and 95th-percentile latency over at least 1,000 runs after warm-up, along with peak memory, average power and energy per inference. State the power mode and numeric precision used.

Submission

Each student submits the following three items for the selected problem statement.

1. **Presentation (PPT, maximum 10 slides)**
   - Problem statement, motivation, contribution, and a brief literature review highlighting the research gap.
   - Dataset description and preprocessing pipeline (EDA, feature engineering, train/validation/test split).
   - Baseline models and the proposed model, with an architecture diagram.
   - Key results: the baseline-versus-proposed comparison table, the ablation study, XAI figures and Jetson measurements.
   - Final conclusion: which approach performs better and why, limitations and future work.
2. **Project report (PDF, maximum 8 pages)**
   - Introduction, problem clarity and contribution.
   - Literature review: existing methods, their limitations and research gaps.
   - Dataset and preprocessing: source, size, statistics and preprocessing steps.
   - Methodology: baselines, the proposed model, mathematical formulation and pseudocode, and a hyperparameter table.
   - Results and discussion: metrics, comparison table, ablation study, XAI analysis, Jetson results, and training curves.
   - Conclusion and references.
3. **Code (.ipynb)**
   - One Jupyter notebook that runs end to end from the public dataset download with a fixed random seed, and reproduces every number in the report.

- Organized into sections: data loading and preprocessing, baseline models, proposed model, training, evaluation, ablation study, XAI and visualization.
- Submitted with all cell outputs saved. Jetson measurement scripts and logs are included as an appendix or a separate script.

File naming: `<RollNo>_MLProject` with the extension `.pptx`, `.pdf` or `.ipynb`.

Deadline and submission link: to be announced by the course coordinator.

Machine Learning Project Evaluation Sheet

S. No.	Evaluation Criteria	Evaluation Parameters	Max. Marks
1	Title, Introduction and contribution	Problem clarity, motivation, contribution	5
2	Literature Review	Existing methods, limitations, research gaps	5
3	Dataset and Preprocessing	Dataset selection, EDA, preprocessing, feature engineering	5
4	Implementation of Proposed model	Algorithm correctness, implementation, hyperparameter tuning	15
5	Proposed Methodology, compare with baseline model, ablation study	Novelty, improvements, mathematical justification	10
6	Results, discussion and Comparative Analysis – table and graph	Performance metrics, baseline comparison, interpretation	10
7	Technical Viva + code demo	Conceptual understanding, mathematics, implementation	30
8	Presentation (10 slides) and Documentation (8 pages max)	Presentation quality, diagram demonstration, code and report	20
	TOTAL		**100**

Datasets and hardware references

Dataset / resource	Used in	Link
NASA C-MAPSS Aircraft Engine Simulator Data	PS 1	data.nasa.gov – C-MAPSS
NASA Li-ion Battery Aging Datasets	PS 2	data.nasa.gov – Li-ion battery ageing
CALCE Battery Data Archive (University of Maryland)	PS 2	calce.umd.edu – battery data
NVIDIA Jetson Orin Nano Super Developer Kit (up to 67 INT8 TOPS, 8 GB LPDDR5, 7–25 W)	Both	NVIDIA user guide; NVIDIA marketplace

Students should cite each dataset as its source page requests, and confirm current download links before starting.